Constraint-Preserving Simulated Bifurcation with Local Degeneracy Aggregation: How Close to Maximum-Likelihood Surface-Code Decoding?

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Abstract

Maximum-likelihood (ML) decoding of a stabilizer code selects the logical class with the largest total probability, whereas Ising-machine decoders usually return a single minimum-weight configuration. We study a decoder that separates the two tasks. A discrete simulated-bifurcation (SB) dynamics searches each logical class in stabilizer-generator coordinates, so that every state it visits is syndrome-feasible, and a local stabilizer-orbit expansion (SOE) around the best candidate of each class estimates the class probability. We benchmark the decoder on the rotated surface code under code-capacity depolarizing noise against exact ML ($d = 3$) and a matrix-product-state (MPS) near-ML reference. After fixing the method on a separate selection set, we evaluate it on a held-out grid of 54 cells ($d = 3$–13, $p = 0.04$–0.18, 7.1×10^6 paired shots). The decoder beats minimum-weight perfect matching and a correlated BP+OSD baseline in all 54 cells after Holm correction. For $d \leq 9$ its logical error rate is within $1.15\times$ of the reference everywhere and statistically indistinguishable from it in 27 of the 36 cells. With the search budget frozen at the value tuned for small codes, accuracy degrades at $d \geq 11$ and the $d = 11/13$ crossing falls to $p \approx 0.11$. Lengthening the bifurcation sweep fourfold reduces the ratio to the reference to 1.07–1.23 and moves the crossing to $p \approx 0.16$ (reference: 0.175). A budget study shows that the sweep length, not the number of replicas, is the resource that must grow with d; that SB and a constraint-preserving simulated-annealing baseline reach the same accuracy plateau, SB with 0.3–$0.6\times$ the search time; and that the remaining excess (at most 13% at the largest budgets) is set by the locality of the degeneracy estimate rather than by the search. On this planar benchmark the MPS decoder is faster at reference accuracy; the proposed decoder requires only a stabilizer generator matrix, and the benchmark measures how far such a generic decoder is from ML.

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Index Terms— Quantum error correction, surface code, maximum-likelihood decoding, degeneracy, simulated bifurcation, Ising machines, simulated annealing.

1 Introduction

A decoder for a stabilizer code receives a syndrome and must return a correction whose product with the actual error is a stabilizer. Because many physical errors share the same syndrome and differ only by stabilizers, the quantity that determines success is the total probability of each *logical class* of errors, not the probability of any single error [1]. Maximum-likelihood decoding (MLD) chooses the class with the largest total probability; minimum-weight decoding chooses the class containing the single most probable error. The two differ precisely when degeneracy matters, and the difference grows with the code distance d and the physical error rate p.

Exact MLD is #P-hard in general [2]. For the planar surface code, tensor-network contraction with a matrix-product-state (MPS) boundary [3] gives a controlled approximation that is essentially exact at moderate bond dimension; generalizations exist for two-dimensional local codes [4, 5]. Monte Carlo methods estimate class probabilities by sampling [6, 7, 8, 9]. Matching [10, 11, 12] and belief propagation with ordered-statistics post-processing (BP+OSD) [13, 14] are fast but ignore part of the degeneracy and, under depolarizing noise, the correlation between X and Z components.

A separate line of work maps decoding onto an Ising problem solved by an annealer or an Ising machine [15, 16, 17, 18]. Writing the error as a fixed syndrome-compatible seed times a product of stabilizer generators makes every spin configuration syndrome-feasible [17, 18]; the objective is then the Pauli weight, a low-order polynomial of the generator spins. These decoders return a minimum-weight solution and therefore, by construction, do not perform MLD.

Simulated bifurcation (SB) [19, 20] is a deterministic, synchronously updated dynamical system for Ising problems. Its update is a matrix–vector-type operation applied to all variables and replicas at once, which makes it attractive for parallel hardware. To our knowledge SB has not

been studied as a quantum decoder.

In this paper we combine SB search in stabilizer coordinates with a cheap estimate of each class's probability mass, and we ask how close the result comes to MLD as the code grows. Our contributions are:

1. A decoder (CP-SB + local SOE, Section 3) that runs discrete SB on the exact Pauli-weight polynomial of each logical class and then aggregates the probability of all errors within a small Hamming radius, in generator space, of the best candidate of each class.
2. A two-stage evaluation (Section 4) in which the method is fixed on a selection set by a rule stated in advance and then tested on a held-out 54-cell grid ($d = 3$–13) against exact MLD and an MPS reference, with paired tests against matching, correlated BP+OSD, and a constraint-preserving simulated-annealing (CP-SA) twin that shares everything but the search dynamics.
3. A decomposition of the remaining gap to ML (Section 5): how much is due to the search (budget), how much to degeneracy that a minimum-weight decision misses, and how much to the locality of the degeneracy estimate. In particular, the sweep length of the bifurcation, not the number of parallel replicas, must grow with d.

We state the limits of the result up front. All experiments use code-capacity noise on the planar rotated surface code, where an MPS decoder gives reference-quality decisions at a cost comparable to ours on a CPU (Section 5.5). We chose this benchmark because it is the setting in which a near-exact reference exists for d up to 13, so that the distance to ML can be measured rather than inferred. The value of the proposed decoder is that its two ingredients need only a stabilizer generator matrix and a Pauli-weight oracle; testing it on codes without planar structure, and on hardware, is left to future work.

2 Background and related work

2.1 Degeneracy and maximum-likelihood decoding

Let $e \in \mathbb{F}_2^{2n}$ denote an n-qubit Pauli error in binary symplectic form and $|e|$ its weight (number of non-identity qubits). Under depolarizing noise each qubit independently suffers X, Y, or Z with probability $p/3$ each, so

$$\Pr(e) \propto w^{|e|}, \qquad w = \frac{p/3}{1-p}. \tag{1}$$

Given the syndrome s, choose any error e_0 with syndrome s, and let ℓ_L ($L = 0, \ldots, 3$) be representatives of the four logical cosets. With $G \in \mathbb{F}_2^{m \times 2n}$ a generator matrix of the stabilizer group, the class-L probability is

$$Z_L(s) = \sum_{u \in \mathbb{F}_2^m} w^{|e_0 \oplus \ell_L \oplus uG|}. \tag{2}$$

MLD returns $\arg\max_L Z_L$; minimum-weight (MAP) decoding returns the class of $\arg\min_{L,u} |e_0 \oplus \ell_L \oplus uG|$. Recent work shows that for planar surface codes the two decisions already differ at error weight $\lceil d/2 \rceil$, i.e., at leading order [21]; degeneracy is therefore not a high-p curiosity on this code.

2.2 Near-ML decoders

The MPS decoder of Bravyi, Suchara and Vargo contracts the planar tensor network of Eq. (2) with bond dimension χ [3]; it is exact as $\chi \to \infty$ and in practice converges at small χ. Markov-chain Monte Carlo over the stabilizer group [6, 7], population annealing [8], worm algorithms [9], and annealed importance sampling with certificates [22] estimate Z_L or ratios of Z_L. Closest to our aggregation step, Liu *et al.* [23] recycle the candidates produced by a stochastic inner decoder into a truncated free energy per class. Our SOE differs in that it does not use the raw candidate pool as the sum: it enumerates a complete Hamming ball in generator space around the single best candidate of each class (Section 3.4), which recovers degenerate neighbors that the solver never visits.

2.3 Ising-machine decoders

Fujisaki *et al.* formulated surface-code decoding as a QUBO with syndrome penalties and solved it on a digital annealer, including circuit-level noise [15, 16]. Takeuchi *et al.* [17] and Takada *et al.* [18] replaced the penalty by stabilizer-generator variables, so that every configuration is feasible, and used simulated annealing [24]. We adopt the same constraint-preserving coordinates; we do not claim them as new. What we add is SB as the search dynamics, an explicit degeneracy-aware decision, and a quantitative comparison with ML.

2.4 Simulated bifurcation

SB integrates a classical Hamiltonian system whose position variables undergo a pitchfork bifurcation as a control parameter is ramped, so that their signs settle into a low-energy Ising configuration [19]. The ballistic and discrete variants (bSB, dSB) add inelastic walls at $|x| = 1$; dSB evaluates the interaction on the signs of the positions [20]. We use a dSB-type update with the exact multilinear force of a higher-order (up to four-body) polynomial.

3 Method

3.1 Code, noise and feasible seed

We use the rotated surface code [25, 26] with $n = d^2$ data qubits and $m = d^2 - 1$ independent stabilizer generators ($(d^2-1)/2$ of each type), and code-capacity depolarizing noise at rate p. A feasible correction e_0 is obtained from

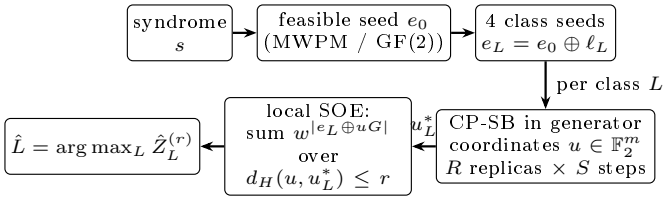


Figure 1: Decoder pipeline. Every state visited by the search is syndrome-feasible because the search runs in stabilizer-generator coordinates. The search supplies the best candidate u_L^* of each class; the local stabilizer-orbit expansion (SOE) supplies an estimate of the class probability Z_L.

minimum-weight perfect matching (MWPM) applied separately to the X and Z components (any GF(2) solution would do); the four class seeds are $e_L = e_0 \oplus \ell_L$ (Fig. 1).

3.2 Pauli-weight Hamiltonian in generator coordinates

For class L every error is $e_L(u) = e_L \oplus uG$ with $u \in \mathbb{F}_2^m$. With spins $\sigma_j = (-1)^{u_j}$, the indicator that qubit q carries a non-identity Pauli is

$$\mathbb{1}[e_q \neq I] = \tfrac{3}{4} - \tfrac{1}{4}\tau_q^x - \tfrac{1}{4}\tau_q^z - \tfrac{1}{4}\tau_q^x\tau_q^z, \tag{3}$$

where $\tau_q^x = (-1)^{(e_L)_q^x} \prod_{j \in \mathcal{X}(q)} \sigma_j$ and $\mathcal{X}(q)$ is the set of generators whose X part acts on q (similarly for z). Summing over q gives the energy $E_L(\sigma) = |e_L(u)|$ exactly, as a multilinear polynomial. On the rotated surface code each qubit is touched by at most two generators of each type, so E_L has terms of order at most four, and there is no penalty term and no auxiliary spin. Because $\Pr(e) \propto w^{|e|}$, minimizing E_L is MAP decoding within class L.

3.3 Constraint-preserving SB (CP-SB)

For each class we evolve R replicas of positions $x \in [-1, 1]^m$ and momenta $y \in \mathbb{R}^m$ for S steps:

$$y \leftarrow y + \Delta t \left[-(1 - a_k)\, x + c\, f(\operatorname{sgn} x)\right], \tag{4}$$

$$x \leftarrow x + \Delta t\, y, \tag{5}$$

followed by inelastic walls ($x_j \leftarrow \operatorname{sgn} x_j$, $y_j \leftarrow 0$ whenever $|x_j| > 1$). The control parameter ramps linearly, $a_k = (k+1)/S$. The force is the exact negative gradient of the multilinear energy at the sign vertex, which we compute from single-generator flip energies:

$$f_j(\sigma) = -\frac{\partial E_L}{\partial \sigma_j} = \tfrac{1}{2}\,\sigma_j \left[E_L(\sigma^{(j)}) - E_L(\sigma)\right], \tag{6}$$

where $\sigma^{(j)}$ is σ with spin j flipped. The coupling is normalized per class, $c = \kappa / \max_j B_j$ with B_j the ℓ_1 norm of the coefficients of all monomials containing j. Each class receives one *warm* restart, initialized near the seed ($x = 0.08 + \mathcal{N}(0, 0.01^2)$, i.e., $u = 0$), and one *random* restart ($x \sim \mathcal{U}(-0.1, 0.1)$), each with R replicas. The terminal sign vectors, together with $u = 0$, form the candidate pool of the class. We denote a budget by $R \times S$; the frozen setting is 8×100 with $\Delta t = 0.45$ and $\kappa = 0.70$, fixed during earlier development on $d \le 9$. A decode therefore runs $4 \times 2 \times R$ trajectories of S steps.

3.4 Local stabilizer-orbit expansion (SOE)

Let u_L^* be the lowest-energy candidate of class L. For radius r we enumerate the Hamming ball $\mathcal{B}_r(u_L^*) = \{u : d_H(u, u_L^*) \le r\}$ in generator space—all products of at most r generators applied to the best candidate—and estimate

$$\hat{Z}_L^{(r)} = \sum_{u \in \mathcal{B}_r(u_L^*)} w^{E_L(u)}, \qquad \hat{L} = \arg\max_L \hat{Z}_L^{(r)}. \tag{7}$$

$\hat{Z}_L^{(r)}$ is a lower bound on Z_L that contains the dominant local degenerate neighbors of the best error. The ball has $1 + m$ states for $r = 1$ and $1 + m + \binom{m}{2}$ for $r = 2$ (169 and 14,197 per class at $d = 13$). SOE needs no sampling and no tuning beyond r; its cost is independent of the search budget. Heavier mass estimators that we tried earlier—adaptive importance sampling and a Bennett acceptance-ratio refinement—did not improve on $r = 2$ on the held-out grid (median $\mathrm{LER}/\mathrm{LER}_{\mathrm{REF}}$ 1.040 and 1.054 versus 1.039) and are not discussed further.

3.5 CP-SA twin

To separate the effect of the search dynamics from everything else, we use a constraint-preserving simulated-annealing twin (CP-SA) that shares the seeds, the Hamiltonian, the pool construction and the SOE step. It performs single-generator Metropolis flips with a geometric temperature schedule from 2.0 to 0.05 over W sweeps, with C chains for each of the warm and random modes; we denote its budget by $C \times W$ (grid setting 8×100).

4 Experimental protocol

4.1 Two-stage design

Selection set. 12,000 shots at $d = 5, 7, 9$ and $p = 0.10, 0.12$ (3000, 2000 and 1000 shots per cell). Before the held-out data existed we fixed the rule: the main decoder is CP-SB + SOE with $r = 1$, replaced by $r = 2$ only if $r = 2$ has strictly fewer failures pooled over the selection set and an exact paired McNemar test gives $p < 0.05$. Radius 2 won (70 shots corrected that $r = 1$ missed versus 40 the other way, $p = 0.0054$), so the main decoder is **CP-SB** 8×100 + **SOE** $r = 2$.

Held-out grid. 54 cells, $d \in \{3, 5, 7, 9, 11, 13\}$, $p \in \{0.04, 0.05, 0.06, 0.08, 0.10, 0.12, 0.14, 0.16, 0.18\}$, with seeds disjoint from the selection set. Shots per cell range from 10^4 ($p \ge 0.14$) to 10^6 ($d \ge 11$, $p = 0.04$), 7.12×10^6 in

total; every decoder decodes the same shots, so all comparisons are paired. The full per-cell table is in Appendix A.

4.2 Baselines

(i) MWPM on the X and Z components with unit weights; PyMatching [11] with p-dependent weights gave identical predictions on every grid shot, and we report them together. (ii) BP+OSD-CS [13, 14] (order 7, min-sum, `ldpc` [27]), first on the X and Z components separately with prior $2p/3$, and (iii) a *correlated* version on the matrix $\left[\begin{smallmatrix} H_Z & 0 & H_Z \\ 0 & H_X & H_X \end{smallmatrix}\right]$ whose three column blocks are the X, Z and Y components, each with prior $p/3$ and $3n$ iterations. The correlated version is better than the independent one in all 54 cells (Holm $p \le 0.0014$; at $d = 13$, $p = 0.10$ its LER/LER$_{\mathrm{REF}}$ is 2.67 instead of 5.14), so it is the BP baseline we report. (iv) CP-SA with the same SOE.

4.3 Reference and its validation

The reference (REF) is exact MLD at $d = 3$ (enumeration of the 2^8 stabilizer group per class), MPS with $\chi = 32$ for $d = 5$–9 [3, 28], and MPS with $\chi = 16$ for $d = 11, 13$, replaced by $\chi = 32$ on every *hard* shot (any decoder wrong or disagreeing with $\chi = 16$; up to 3000 per cell). Checks: at $d = 3$ the MPS decisions agree with exact MLD on 100% of non-tied syndromes (4695 and 4610 shots at $p = 0.10, 0.12$). On the grid, $\chi = 16$ and $\chi = 32$ disagree on at most 7 shots per cell at $d = 9$ and at most 2 hard shots per cell at $d \ge 11$; on the $d \ge 11$ hard shots $\chi = 32$ and $\chi = 64$ disagree on at most 1 shot per cell (18 cells, 401–3000 shots each). At $d = 5$ the $\chi = 16/32$ disagreements are more frequent (67–188 per cell), but they are near-ties and vanish at larger d; we use $\chi = 32$ there.

4.4 Metrics and statistics

We report the logical error rate (LER) with Wilson 95% intervals, the ratio LER/LER$_{\mathrm{REF}}$, and the gap closure relative to MWPM,

$$G = \frac{\mathrm{LER}_{\mathrm{MWPM}} - \mathrm{LER}_X}{\mathrm{LER}_{\mathrm{MWPM}} - \mathrm{LER}_{\mathrm{REF}}}, \tag{8}$$

with a paired bootstrap interval. Paired comparisons use the exact two-sided McNemar (sign) test on discordant shots, Holm-corrected across all cells of a comparison family. Where per-cell tests are underpowered we also report the discordant counts pooled over cells with an exact sign test; we treat these pooled tests as descriptive.

4.5 Budget study and budget-scaled rows

On the first 10^4 shots of every cell with $d \in \{9, 11, 13\}$ and $p = 0.08$–0.16 we decoded with CP-SB budgets $R \times S \in \{8{\times}100, 16{\times}100, 32{\times}100, 64{\times}100, 8{\times}200, 8{\times}400, 32{\times}400\}$ and CP-SA budgets $C \times W \in \{8 \times 25, 8 \times 50, 8 \times 100, 8 \times 200, 16 \times 100, 32 \times 200\}$, each with MAP, SOE $r = 1$ and SOE $r = 2$ decisions; at $p = 0.18$ we ran five of these configurations. The 8×100 configurations reproduce the grid decisions shot by shot. From this study we then fixed a single rule—keep $R = 8$ and use $S = 400$ for $d \ge 11$—and applied it to the full grid cells at $d = 11, 13$ (all shots for $p \ge 0.08$, the first 10^5 shots for $p \le 0.06$). Because the rule was chosen after the budget study, and at $p \ge 0.14$ the study used the same shots, we present these *budget-scaled* rows as a budget study next to the frozen held-out rows, not as an independent confirmation.

Table 1: LER/LER$_{\mathrm{REF}}$ on the held-out grid (selected cells; all 54 in Appendix A). REF in percent. *: significantly worse than REF (Holm $p < 0.05$). The last column is the budget-scaled 8×400 row (Section 5.3).

d	p	REF (%)	MWPM	BP+OSD$_c$	SA+SOE$_1$	SB+SOE$_2$	SB$_{400}$+SOE$_2$
3	0.08	6.99	1.13	1.05	1.00	1.01	–
3	0.12	13.8	1.11	1.05	1.00	1.01	–
3	0.16	21.8	1.09	1.04	0.99	0.99	–
5	0.08	3.74	1.51	1.23	1.02	1.02	–
5	0.12	10.9	1.34	1.16	1.03	1.01	–
5	0.16	20.8	1.25	1.13	1.02	1.00	–
7	0.08	1.94	2.04	1.53	1.04	1.04	–
7	0.12	7.74	1.71	1.44	1.07	1.06*	–
7	0.16	18.7	1.46	1.34	1.07	1.06*	–
9	0.08	0.94	2.82	2.00	1.03	1.01	–
9	0.12	6.08	2.02	1.66	1.09	1.08*	–
9	0.16	17.3	1.62	1.46	1.09	1.10*	–
11	0.08	0.50	3.97	2.68	1.07	1.20*	1.10*
11	0.12	4.28	2.55	2.18	1.14	1.28*	1.12*
11	0.16	16.3	1.73	1.60	1.11	1.22*	1.10*
13	0.08	0.23	5.74	3.99	1.23	1.87*	1.07
13	0.12	3.12	3.28	2.65	1.25	1.82*	1.18*
13	0.16	15.1	1.93	1.88	1.25	1.54*	1.19*

4.6 Timing

Times are single-process CPU wall times per unique nontrivial syndrome on an otherwise idle node ([TODO: CPU model]; Python 3.11, NumPy 2.4), medians over 100 unique syndromes at $p = 0.10$ for $d \ge 7$, for every decoder and budget configuration. A second timing run at $p = 0.14$ gives medians within a few percent of those at $p = 0.10$ for every configuration. Our implementation is unoptimized NumPy; PyMatching is compiled C++; the MPS decoder is the Python implementation in `qecsim` [28]. Times therefore describe this benchmark, not hardware limits.

5 Results

5.1 Comparison with matching and BP+OSD

The main decoder has a lower LER than MWPM/PyMatching and than correlated BP+OSD in all 54 cells, each significant after Holm correction (largest Holm p: 8.7×10^{-12} against MWPM, 0.0077 against correlated BP+OSD). Pooled over the grid, it corrects 55,081 shots that MWPM fails on versus 17,007 in the opposite direction, and 41,739 versus 21,099 against correlated BP+OSD. The margins grow with d

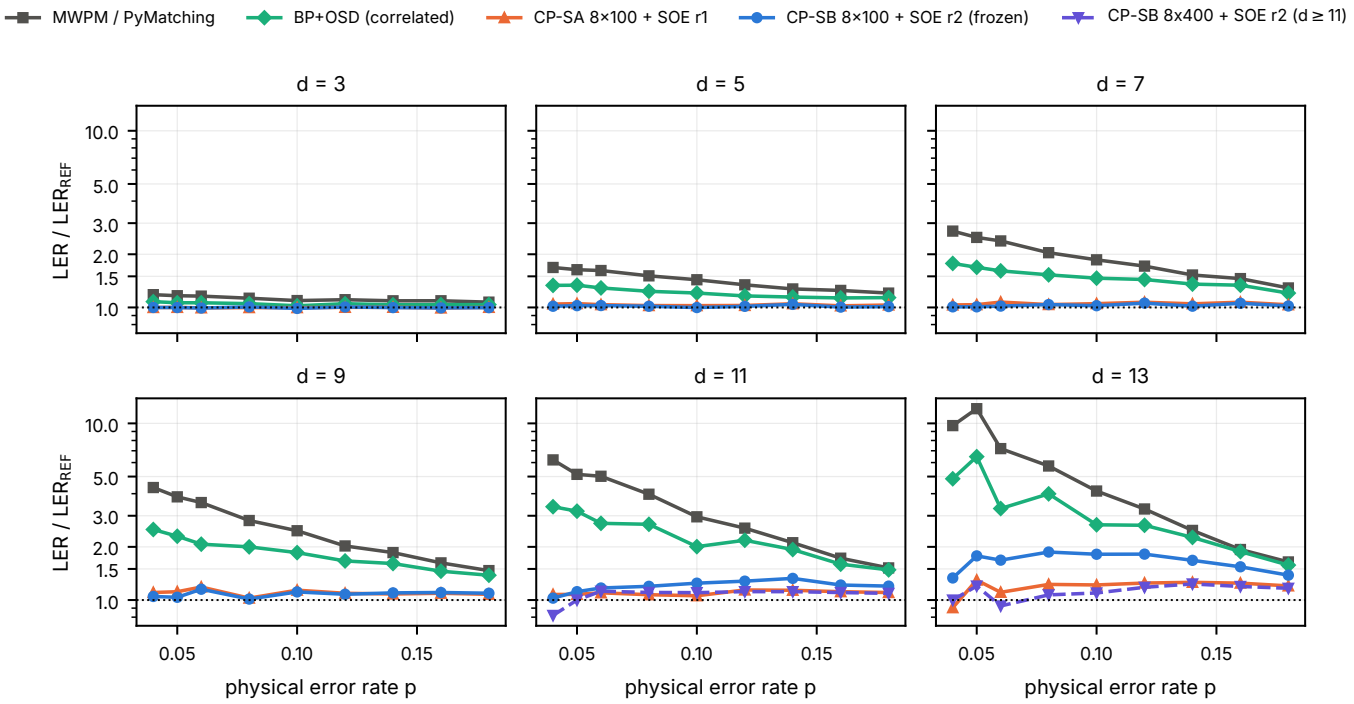


Figure 2: Logical error rate relative to the reference on the held-out grid. Dotted line: reference (exact MLD at $d = 3$, MPS otherwise). Dashed: the budget-scaled CP-SB 8×400 + SOE $r = 2$ at $d \geq 11$ (Section 5.3). At $d = 11, 13$ and $p \leq 0.06$ the reference has fewer than 100 failures per cell and the ratios are noisy.

(Fig. 2, Table 1): at $d = 13$, MWPM is 1.6–12× and correlated BP+OSD 1.6–6.5× the reference, compared with 1.3–1.9× for our frozen decoder. Beating MWPM under depolarizing noise is expected of any decoder that uses the X–Z correlation; the correlated BP+OSD comparison is the more informative one. No decoder lost effective distance: across the grid there is no failure of weight below $\lceil d/2 \rceil$ except for independent BP+OSD (1–3 failures per cell at $d = 9, 11$, $p \leq 0.06$).

5.2 Gap to ML at fixed budget

For $d \leq 9$ the frozen decoder is close to the reference: $\mathrm{LER}/\mathrm{LER}_{\mathrm{REF}}$ lies in 0.99–1.01 at $d = 3$, 1.00–1.04 at $d = 5$, 1.01–1.06 at $d = 7$ and 1.01–1.15 at $d = 9$; the gap closure is at least 0.85 for $d \leq 7$ and at least 0.80 at $d = 9$. Across all 54 cells the paired test against the reference is not significant in 30 cells; the 24 significant cells (all in the direction of a higher LER for our decoder) are distributed as $0, 1, 2, 6, 7, 8$ over $d = 3, \ldots, 13$. At $d = 11, 13$ the frozen decoder degrades ($\mathrm{LER}/\mathrm{LER}_{\mathrm{REF}}$ up to 1.33 and 1.87), while the CP-SA twin at the same nominal budget stays at 1.06–1.14 and 0.90–1.30; CP-SA + SOE $r = 1$ is significantly better than CP-SB + SOE $r = 1$ in 15 cells, all of them at $d = 11$ ($p \geq 0.06$) or $d = 13$ ($p \geq 0.05$), and never the reverse. As the next subsection shows, this difference is a matter of search time: on an idle core, the frozen SB search at $d = 13$ takes 0.45 s per syndrome and the CP-SA search 2.77 s.

The frozen budget also suppresses the threshold behavior. With 8×100, $\mathrm{LER}(d = 13)$ exceeds $\mathrm{LER}(d = 11)$ from $p = 0.12$ on (5.69% vs 5.48%), a crossing at $p \approx 0.11$; the reference curves for $d = 11$ and 13 cross at $p \approx 0.175$ (Fig. 3).

5.3 Scaling the search budget

Budget-scaled rows. Raising the number of SB steps from 100 to 400 at $d \geq 11$ (replicas unchanged) lowers the LER in all 18 cells relative to the frozen decoder, significantly in 12 (Holm), all at $d = 11$, $p \geq 0.10$ or $d = 13$, $p \geq 0.06$; pooled, it corrects 6,429 shots that the frozen decoder fails on versus 3,487. For $p \geq 0.08$, $\mathrm{LER}/\mathrm{LER}_{\mathrm{REF}}$ falls from 1.20–1.33 to 1.09–1.12 at $d = 11$ and from 1.38–1.87 to 1.07–1.23 at $d = 13$ (Table 1, dashed lines in Fig. 2). The budget-scaled rows remain significantly worse than the reference in 10 of 18 cells ($d = 11$, $p \geq 0.08$; $d = 13$, $p \geq 0.12$), by 9–23%.

Threshold. With 8×400 the $d = 11/13$ crossing moves from $p \approx 0.11$ to $p \approx 0.16$ (LER at $p = 0.16$: 17.95% vs 18.07%). In the budget study, CP-SB 32×400 and CP-SA 32×200 give LERs that decrease monotonically with d up to $p = 0.16$ (CP-SB 32×400 at $p = 0.16$: 18.68, 17.68, 16.72% for $d = 9, 11, 13$) and cross between 0.16 and 0.18 ($p \approx 0.17$), approaching the reference value ≈ 0.175. The low crossing of the frozen decoder is therefore a property of a budget that does not grow with d—CP-SA 8×100 also crosses early, between $p = 0.14$ and 0.16—and not a limit of the method. These finite-size crossings lie below the asymptotic optimal threshold of ≈ 0.189 [29], as expected

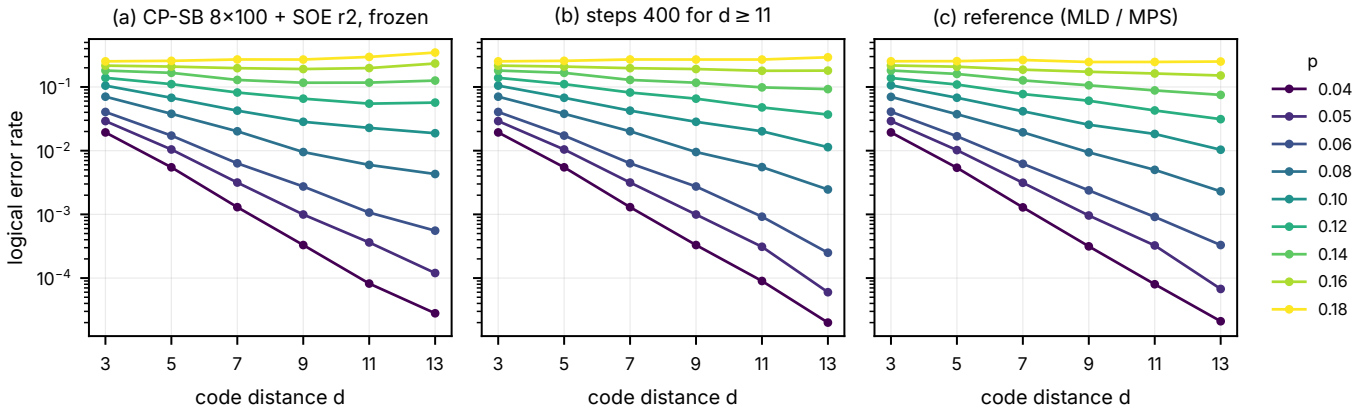


Figure 3: Logical error rate versus d for each p. (a) Frozen budget CP-SB 8×100 + SOE $r = 2$ at every d. (b) Same decoder with $S = 400$ steps for $d \geq 11$. (c) Reference. With the frozen budget the curves turn upward at $d = 13$ already at $p \approx 0.12$; with the longer sweep they decrease with d up to $p = 0.14$.

at $d \leq 13$.

Steps versus replicas. At equal search time the longer sweep is better. At $d = 13$, CP-SB 8×400 (1.74 s) and 32×100 (1.66 s) give $\mathrm{LER}/\mathrm{LER}_{\mathrm{REF}}$ (SOE $r = 2$, $p = 0.10$–0.16) of 1.10, 1.17, 1.23, 1.19 and 1.23, 1.28, 1.33, 1.29; pooled over the 15 study cells, 8×400 corrects 1,935 shots that 32×100 fails on versus 1,643 (sign test $p \approx 10^{-6}$; one cell significant on its own). 64×100, at about twice the time (3.28 s), only matches 8×400 (1,469 vs 1,500). Replicas are independent samples of the same quench: if the bifurcation is traversed too quickly for a given m, more replicas reproduce the same bias. A longer ramp lets the positions follow the slow bifurcation and settle into lower-energy sign patterns (Fig. 4, top).

SB versus SA. When the decision rule is held fixed, the two searches converge to the same accuracy. Comparing budgets of similar accuracy, CP-SB 8×400 versus CP-SA 8×100 and CP-SB 32×400 versus CP-SA 32×200, none of the 18 cells is significant for any of MAP, SOE $r = 1$ or SOE $r = 2$, and the pooled discordant counts are balanced (SOE $r = 2$: 3,169 vs 3,048 and 1,134 vs 1,200). The SB searches take 1.74 and 6.55 s at $d = 13$, versus 2.77 and 22.0 s for SA, i.e., 0.63× and 0.30×. The budget-scaled main decoder (8×400, $r = 2$) is not significantly different from the grid's CP-SA 8×100 + SOE $r = 1$ in any of the 18 full cells either (pooled 3,265 vs 2,914, a small edge for SB). At low budget SB is clearly better: CP-SB 8×100 against CP-SA 8×25 (search times 0.45 and 0.71 s at $d = 13$) is significant in 8 of 15 cells for SOE $r = 1$, pooled 6,124 vs 3,867. We therefore claim the same accuracy plateau at a smaller search time, not a better plateau.

5.4 Where the remaining gap comes from

Figure 5 decomposes the gap into search and degeneracy. The MAP decision (lowest-weight candidate over all classes) is 1.04–1.12× the reference at $d = 3$ and rises to 1.2–1.5× at $d = 9$ and 1.4–2.7× at $d = 13$ with the frozen budget. With a sufficient budget all searches return the same minimum-weight decision: at $d = 13$, $p = 0.12$ the MAP ratio is 1.37–1.38 for CP-SA 8×200, 16×100, 32×200 and CP-SB 32×400. The remaining MAP excess is therefore not a search failure but degeneracy: the most probable single error lies in the wrong class. SOE recovers most of it—in the same cell SOE $r = 2$ lowers the ratio from 1.38 to 1.10—and the share it recovers grows with d.

Radius 2 is better than radius 1 by a small, consistent margin. On the grid only 3 of 54 cells are individually significant, but 50 of 54 point the same way, and pooled over the grid $r = 2$ corrects 5,222 shots that $r = 1$ misses versus 4,337 ($p = 1.5 \times 10^{-19}$). At $d = 3$, where $r = 1$ is already essentially MLD, the two are tied (1,893 vs 1,882); every $d \geq 5$ is significant on its own ($p < 10^{-3}$). With the 8×400 search the picture is the same (0/18 cells, 12/18 in favor, pooled 569 vs 376).

At the largest budgets the ratio to the reference levels off independently of the search: SOE $r = 2$ gives 1.05–1.08 at $d = 9$, 1.00–1.11 at $d = 11$ and 1.08–1.13 at $d = 13$ for $p \geq 0.10$, for both CP-SB 32×400 and CP-SA 32×200. This plateau is the error of the local estimate (7) itself: the ball around a single center misses degenerate errors that differ from it by larger stabilizer products and the mass around other near-optimal centers. More search does not remove it; a larger or adaptive orbit would.

5.5 Cost

Table 2 lists per-syndrome times. Two points matter for interpreting the results. First, at $d \geq 11$ the SOE $r = 2$ step, whose cost grows as m^2, is more expensive than the search: at $d = 13$ it takes 3.5 s against 1.7 s for the 8×400 search. Once the search is adequate, $r = 2$ buys a 1–3% relative improvement over $r = 1$ at $d = 13$ for about 3.5 s, whereas the same time spent on SB steps buys much more (Fig. 4, bottom). The cost-effective configuration at $d \geq 11$ is therefore a longer SB sweep with $r = 1$ (1.8 s end to end at $d = 13$); $r = 2$ remains the accuracy headline

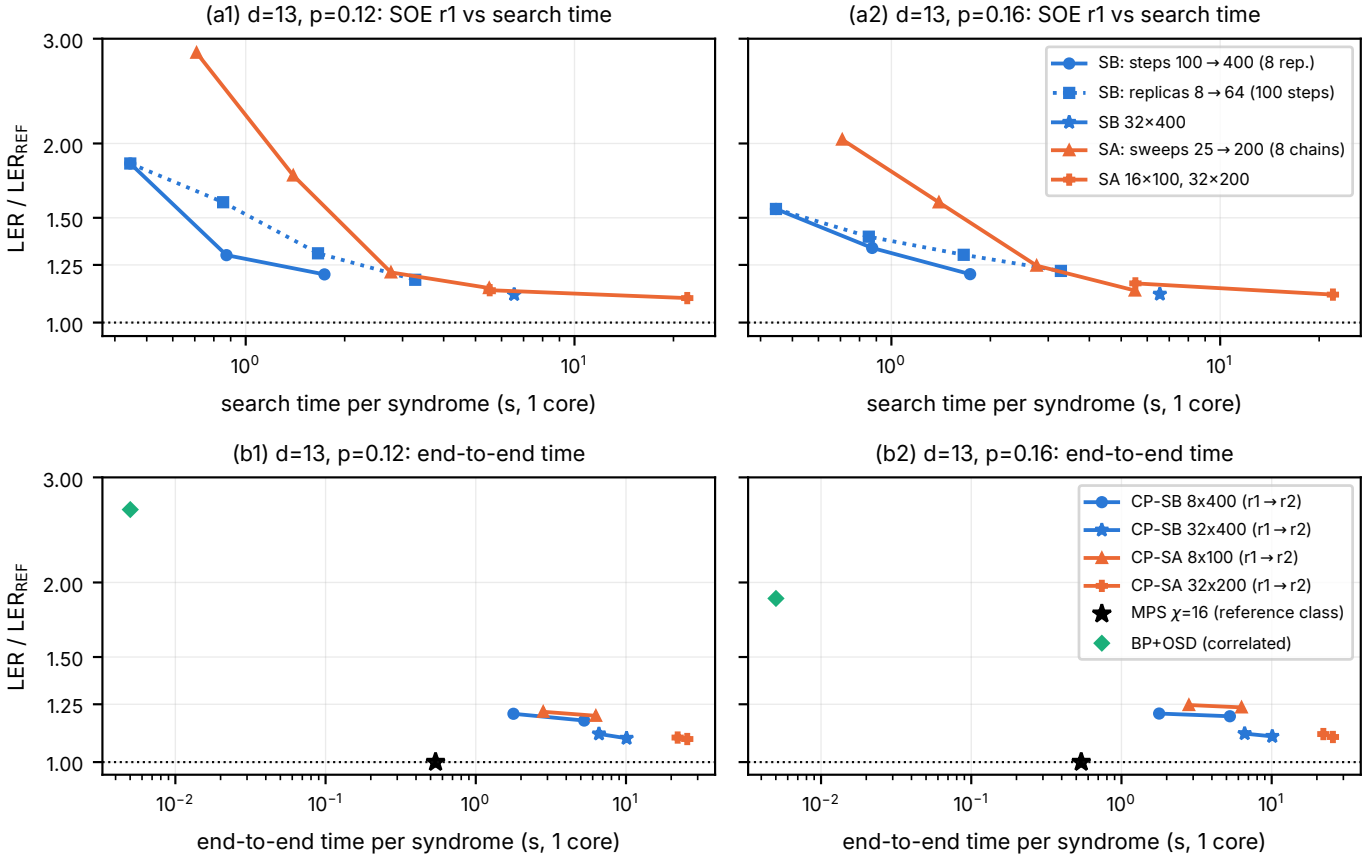


Figure 4: Budget study at $d = 13$. Top: SOE $r = 1$ accuracy versus search time per syndrome for CP-SB (blue) and CP-SA (orange); solid lines increase the number of steps (sweeps), dotted lines the number of replicas. Bottom: accuracy versus end-to-end time (seed + search + SOE) with $r = 1$ and $r = 2$ joined by a line; star: MPS $\chi = 16$; diamond: correlated BP+OSD. Times are from the budget-study timing run at $p = 0.10$.

Table 2: Median single-core time per unique non-trivial syndrome at $p = 0.10$ (seconds). 100 syndromes per entry. Unoptimized Python/NumPy except PyMatching (C++) and BP+OSD (Cython). End-to-end times include the MWPM seed.

	$d = 9$	$d = 11$	$d = 13$
PyMatching	2.7×10^{-5}	3.3×10^{-5}	4.3×10^{-5}
BP+OSD, correlated	0.0011	0.0025	0.0050
MPS $\chi = 16$	0.164	0.326	0.540
MPS $\chi = 32$	0.659	1.711	3.269
CP-SB search 8×100	0.133	0.250	0.446
CP-SB search 8×400	0.511	0.968	1.736
CP-SB search 32×400	1.768	3.552	6.554
CP-SA search 8×100	1.317	1.980	2.765
CP-SA search 32×200	10.49	15.68	22.02
SOE $r = 1$	0.007	0.017	0.040
SOE $r = 2$	0.258	1.061	3.524
End to end: 8×100 + SOE $r = 2$ (frozen)	0.393	1.316	3.980
End to end: 8×400 + SOE $r = 1$	0.520	0.989	1.781
End to end: 8×400 + SOE $r = 2$	0.771	2.033	5.267

because it was selected in advance.

Second, on this benchmark the MPS decoder with $\chi = 16$ reaches reference accuracy in 0.54 s at $d = 13$, about the time of our frozen search alone and 3–10× faster than our budget-scaled pipeline (1.78 s with $r = 1$, 5.27 s with $r = 2$); even with $\chi = 32$ it takes 3.3 s. Our decoder does not dominate MPS on a planar code in accuracy or in CPU time, and we do not claim it does. MPS contraction relies on the planar two-dimensional structure of the code; the two components of our decoder use only the generator matrix G and the Pauli weight, and the dynamics is a synchronous update of all replicas and variables suited to parallel hardware. Neither of these properties is tested here (Section 6).

5.6 Reproducibility across solver seeds

Three independent solver seeds for CP-SB 8×100 and CP-SA 8×100 (SOE $r = 1$; $d = 9, 11, 13$; $p = 0.10, 0.14$; 10^4 shots each) give LERs within 0.06–0.54 percentage points of each other. The fraction of shots on which all three seeds agree is 0.87–0.99 for CP-SB and 0.90–0.99 for CP-SA; the lower agreement of the frozen SB search at $d = 13$, $p = 0.14$ (0.865 vs 0.905) is another sign that 8×100 does not converge there.

6 Discussion

What the benchmark shows. On the one code family where near-exact ML decisions are available up to $d = 13$, a de-

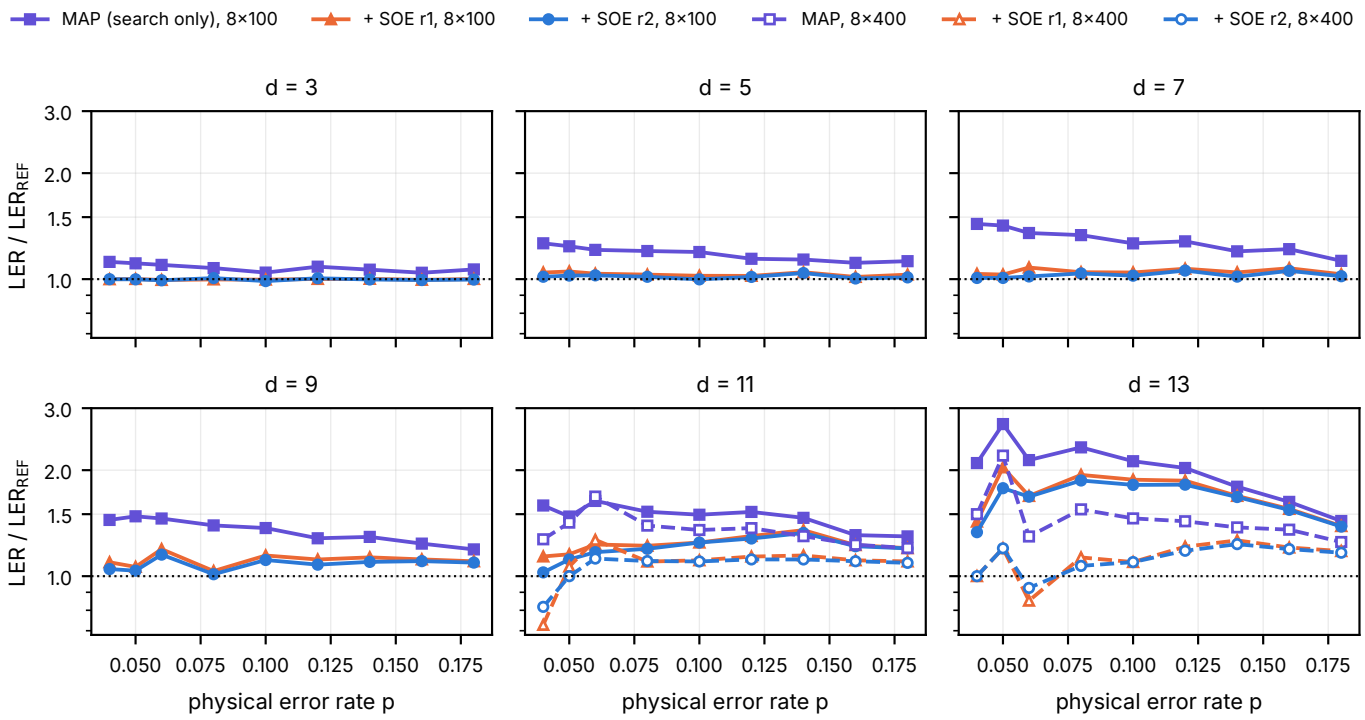


Figure 5: Gap decomposition on the held-out grid: minimum-weight decision of the search (MAP), plus SOE $r = 1$, plus SOE $r = 2$, all relative to the reference. Solid: frozen 8×100 search. Dashed, open markers ($d \geq 11$): 8×400 search. The distance between MAP and SOE is the degeneracy that a minimum-weight decoder misses; the distance between SOE $r = 2$ and 1 is the error of the local mass estimate.

coder made of a generic Ising search and a local degeneracy estimate comes within 1.15× of ML for $d \leq 9$ and within 1.07–1.23× at $d = 11, 13$ once its search budget grows with d. It beats correlated BP+OSD everywhere. The remaining excess separates cleanly into three parts: an undersized search budget (removed by longer SB sweeps), degeneracy that a minimum-weight decision misses (mostly recovered by SOE), and the locality of SOE (a plateau of up to 13% that no amount of search removes).

The resource that must scale. The number of SB steps, not the number of replicas, has to grow with d. In stabilizer coordinates the energy landscape is a frustrated, up-to-four-body Ising model on $m = d^2 - 1$ spins whose low-energy states are separated by stabilizer moves of growing size; a ramp that is fast compared with the slowest bifurcating mode leaves every replica with the same residual defects. A practical rule is to keep a small replica count and lengthen the sweep. For SA the analogous resource is the number of sweeps; at equal accuracy SB needs 0.3–0.6× the search time of SA in our implementation.

Relation to MPS and to sampling decoders. For the planar surface code an MPS decoder is the natural choice: it is near-exact at modest χ and, in our CPU implementation, faster. Our results should be read as a measurement of how much accuracy a code-agnostic optimize-then-aggregate decoder loses relative to ML, and of where that loss comes from. Sampling estimators of Z_L [7, 8, 9, 22] and truncated free energies [23] target the same quantity; local SOE is the cheapest such estimate, needs no tuning, and, as Section 5.4 shows, already accounts for most of the degeneracy on this code.

Beyond code capacity. The constraint-preserving coordinates need a generating set of the group of fault configurations that leave every detector and logical observable unchanged. Under circuit-level noise this group is the kernel of the stacked detector and observable matrices of the detector error model (DEM), and the question is whether it has a *local* generating set, so that the energy remains a low-order polynomial. We checked this structural precondition, without decoding, for rotated surface-code memory experiments with $d = 3$–13 and d rounds and for the $[[72, 12, 6]]$ bivariate-bicycle (BB) code with its depth-8 syndrome cycle [30] and 6 rounds, at $p = 0.003$ (Appendix C). Generators are found by decoding each fault mechanism's own detector pattern with that mechanism excluded, followed by a greedy GF(2) basis; any rank deficit is filled from the global null space and reported separately. For the surface code the self-decoded generators have weight 3 at the median and at most 4 at the 99th percentile for every d, and only $0, 4, 6, 6, 13, 14$ completion elements are needed for $d = 3, \ldots, 13$; at $d = 13$, 11 of the 14 are time-like elements that run through (nearly) all rounds, with weights up to 57, among 39,724 generators. Each fault mechanism is acted on by 2.9 generators on average (≤ 16 at the 99th percentile), which is the order of the corresponding energy term. For the BB code the

self-decoded generators are equally local (99th-percentile weight 4), but 261 completion elements (1.7%, weights up to 104, only greedily lightened) are needed. The search space thus stays local at circuit level apart from a small set of global elements, which grows slowly with d for the surface code and must be treated separately; we have not evaluated decoding accuracy in this setting.

Limitations. (i) Decoding accuracy is measured for code-capacity depolarizing noise on the planar rotated surface code only; for circuit-level noise we verify only the structural precondition above. (ii) The budget-scaled rows use a rule chosen after the budget study on overlapping shots (Section 4.5); the frozen rows are the held-out result. (iii) At $d \geq 11$ and $p \leq 0.06$ the reference has few failures per cell (2–82 on the decoded subsets), so the ratios there are imprecise. (iv) The reference at $d \geq 5$ is MPS, not exact MLD; the convergence checks above bound, but do not eliminate, its error. (v) Timing is single-core Python and should not be used to compare against optimized or hardware implementations. (vi) We did not test codes without planar structure.

Outlook. Three directions follow directly from the decomposition: an SOE whose radius or set of centers adapts to the ambiguity of the decision, which targets the plateau; a parallel implementation of the synchronous SB update, which targets the search time; and circuit-level noise or codes for which no tensor-network reference is efficient, where the generic formulation is the point and where the global elements found in Appendix C must be handled explicitly.

7 Conclusion

Constraint-preserving simulated bifurcation in stabilizer coordinates, followed by a local stabilizer-orbit estimate of each logical class's probability, decodes the rotated surface code under code-capacity depolarizing noise at near-ML accuracy for $d \leq 9$ and within 7–23% of an MPS reference at $d = 11, 13$ when the bifurcation sweep is lengthened with d. It outperforms matching and correlated BP+OSD in all 54 held-out cells. The budget study identifies the sweep length as the resource that must scale, shows that SB and SA share the same accuracy plateau with SB reaching it in less search time, and attributes the residual gap to the locality of the degeneracy estimate.

Data and code availability

All code (decoders, experiment drivers, the circuit-level locality analysis, Slurm scripts and analysis), the per-cell summary tables and the figure scripts are available at [TODO: repository / Zenodo DOI]. The per-shot outputs (≈1.5 GB) are available on request [TODO: or Zenodo].

A Full held-out grid

Table 3 lists every cell. Columns as in Table 1; additionally the minimum-weight decision of the frozen search (SB MAP), SOE $r = 1$ (SB+SOE$_1$), and the gap closure G of the main decoder. For the budget-scaled column at $p \leq 0.06$ the ratio is computed on the decoded subset (10^5 shots).

B Budget study at $d = 13$

At $p = 0.08$ the reference has only 16 failures in 10^4 shots, so that column is coarse.

C Locality of the circuit-level stabilizer group

Circuits: Stim [31] `rotated_memory_z` for the surface code with `after_clifford_depolarization`, `before_round_data_depolarization`, `before_measure_flip_probability` and `after_reset_flip_probability` all equal to p; for the BB code, the depth-8 cycle of Ref. [30] with two-qubit depolarizing noise after every CNOT, single-qubit depolarizing noise on idle data qubits in every layer and on data qubits before each round, and flips after reset and before measurement, all with probability $p = 0.003$. Self-decoding uses BP+OSD (order 7) on the stacked detector–observable matrix with three weight profiles per mechanism. Completion elements of the surface code are lightened by a randomized local information-set search; for the BB code only greedy pairwise lightening was used, so their weights are upper bounds. In Table 5, weight is the number of fault mechanisms in a self-decoded generator, "global" is the number of completion elements and their weight range, and "order" is the number of generators acting on one mechanism.

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Table 3: All 54 held-out cells: shots, reference LER (%), and $\mathrm{LER}/\mathrm{LER}_{\mathrm{REF}}$ of each decoder. *: main decoder significantly worse than REF (Holm $p < 0.05$).

d	p	shots	REF (%)	MWPM	BP+OSD$_{\mathrm{c}}$	SA+SOE$_1$	SB MAP	SB+SOE$_1$	SB+SOE$_2$	G	SB$_{400}$+SOE$_2$
3	0.04	400,000	1.93	1.18	1.08	1.00	1.12	1.00	1.00	1.00	–
3	0.05	200,000	2.92	1.17	1.06	1.00	1.11	1.00	1.00	1.01	–
3	0.06	100,000	4.08	1.16	1.06	0.99	1.10	0.99	0.99	1.05	–
3	0.08	50,000	6.99	1.13	1.05	1.00	1.07	1.00	1.01	0.95	–
3	0.10	20,000	10.6	1.09	1.02	1.00	1.04	1.00	0.99	1.14	–
3	0.12	20,000	13.8	1.11	1.05	1.00	1.08	1.00	1.01	0.95	–
3	0.14	10,000	18.1	1.09	1.04	1.00	1.06	1.00	1.00	1.02	–
3	0.16	10,000	21.8	1.09	1.04	0.99	1.04	1.00	0.99	1.07	–
3	0.18	10,000	25.3	1.07	1.04	1.00	1.07	1.00	1.00	1.04	–
5	0.04	400,000	0.539	1.68	1.33	1.04	1.26	1.04	1.02	0.98	–
5	0.05	200,000	1.02	1.64	1.34	1.05	1.24	1.05	1.02	0.96	–
5	0.06	100,000	1.68	1.62	1.29	1.04	1.21	1.04	1.02	0.96	–
5	0.08	50,000	3.74	1.51	1.23	1.02	1.20	1.03	1.02	0.97	–
5	0.10	20,000	6.78	1.43	1.21	1.02	1.19	1.02	1.00	1.00	–
5	0.12	20,000	10.9	1.34	1.16	1.03	1.14	1.02	1.01	0.96	–
5	0.14	10,000	16	1.27	1.14	1.05	1.14	1.05	1.04*	0.85	–
5	0.16	10,000	20.8	1.25	1.13	1.02	1.11	1.01	1.00	0.98	–
5	0.18	10,000	25.4	1.21	1.14	1.03	1.12	1.03	1.01	0.94	–
7	0.04	400,000	0.129	2.70	1.77	1.03	1.43	1.03	1.01	1.00	–
7	0.05	200,000	0.314	2.49	1.69	1.04	1.42	1.03	1.01	0.99	–
7	0.06	100,000	0.622	2.38	1.61	1.07	1.35	1.08	1.02	0.99	–
7	0.08	50,000	1.94	2.04	1.53	1.04	1.33	1.05	1.04	0.96	–
7	0.10	20,000	4.15	1.86	1.46	1.05	1.26	1.04	1.02	0.97	–
7	0.12	20,000	7.74	1.71	1.44	1.07	1.28	1.07	1.06*	0.92	–
7	0.14	10,000	12.7	1.53	1.36	1.05	1.20	1.04	1.02	0.97	–
7	0.16	10,000	18.7	1.46	1.34	1.07	1.22	1.07	1.06*	0.88	–
7	0.18	10,000	26.5	1.29	1.21	1.04	1.13	1.03	1.02	0.93	–
9	0.04	400,000	0.0315	4.33	2.51	1.10	1.44	1.10	1.05	0.99	–
9	0.05	200,000	0.096	3.84	2.30	1.11	1.48	1.06	1.04	0.99	–
9	0.06	100,000	0.238	3.56	2.07	1.18	1.46	1.19	1.15*	0.94	–
9	0.08	50,000	0.94	2.82	2.00	1.03	1.39	1.03	1.01	0.99	–
9	0.10	20,000	2.56	2.47	1.86	1.14	1.37	1.14	1.11*	0.92	–
9	0.12	20,000	6.08	2.02	1.66	1.09	1.28	1.12	1.08*	0.92	–
9	0.14	10,000	10.6	1.86	1.61	1.08	1.29	1.13	1.10*	0.89	–
9	0.16	10,000	17.3	1.62	1.46	1.09	1.24	1.12	1.10*	0.83	–
9	0.18	10,000	24.7	1.47	1.38	1.08	1.19	1.10	1.09*	0.80	–
11	0.04	1,000,000	0.008	6.21	3.37	1.07	1.59	1.14	1.02	1.00	0.82
11	0.05	400,000	0.0325	5.15	3.18	1.08	1.48	1.15	1.12	0.97	1.00
11	0.06	400,000	0.0912	5.02	2.72	1.10	1.64	1.23	1.17*	0.96	1.12
11	0.08	50,000	0.5	3.97	2.68	1.07	1.52	1.22	1.20*	0.93	1.10
11	0.10	20,000	1.83	2.96	2.01	1.06	1.49	1.25	1.25*	0.87	1.10
11	0.12	20,000	4.28	2.55	2.18	1.14	1.52	1.30	1.28*	0.82	1.12
11	0.14	10,000	8.83	2.11	1.93	1.14	1.46	1.35	1.33*	0.71	1.12
11	0.16	10,000	16.3	1.73	1.60	1.11	1.31	1.22	1.22*	0.70	1.10
11	0.18	10,000	24.7	1.52	1.48	1.10	1.30	1.20	1.20*	0.62	1.09
13	0.04	1,000,000	0.0021	9.71	4.86	0.90	2.10	1.43	1.33	0.96	1.00
13	0.05	400,000	0.00675	12.11	6.48	1.30	2.70	2.04	1.78*	0.93	1.20
13	0.06	400,000	0.033	7.20	3.30	1.11	2.14	1.69	1.68*	0.89	0.93
13	0.08	50,000	0.23	5.74	3.99	1.23	2.32	1.94	1.87*	0.82	1.07
13	0.10	20,000	1.03	4.14	2.67	1.22	2.12	1.88	1.82*	0.74	1.10
13	0.12	20,000	3.12	3.28	2.65	1.25	2.03	1.87	1.82*	0.64	1.18
13	0.14	10,000	7.51	2.48	2.26	1.26	1.79	1.69	1.68*	0.54	1.23
13	0.16	10,000	15.1	1.93	1.88	1.25	1.63	1.55	1.54*	0.42	1.19
13	0.18	10,000	25.1	1.64	1.58	1.20	1.43	1.39	1.38*	0.40	1.17

Table 4: $\mathrm{LER}/\mathrm{LER}_{\mathrm{REF}}$ (SOE $r = 2$) at $d = 13$ on the first 10^4 shots of each cell, and search time per syndrome (s, $p = 0.10$).

budget	time	0.08	0.10	0.12	0.14	0.16	0.18
SB 8×100	0.45	2.50	1.84	1.82	1.68	1.54	1.38
SB 8×200	0.87	1.25	1.37	1.29	1.40	1.32	–
SB 8×400	1.74	1.31	1.10	1.17	1.23	1.19	1.17
SB 16×100	0.85	1.88	1.49	1.55	1.46	1.39	–
SB 32×100	1.66	1.63	1.23	1.28	1.33	1.29	–
SB 64×100	3.28	1.50	1.12	1.17	1.23	1.21	–
SB 32×400	6.55	1.19	1.11	1.10	1.13	1.10	1.08
SA 8×25	0.71	3.88	2.91	2.60	2.27	1.96	–
SA 8×50	1.39	1.75	1.69	1.77	1.69	1.58	–
SA 8×100	2.77	1.19	1.21	1.20	1.25	1.24	1.20
SA 8×200	5.50	1.19	1.06	1.11	1.12	1.12	–
SA 16×100	5.52	1.19	1.15	1.11	1.16	1.15	–
SA 32×200	22.0	1.19	1.11	1.09	1.12	1.10	1.08

Table 5: Circuit-level DEM stabilizer group at $p = 0.003$: mechanisms, stabilizer dimension (spins), weight of self-decoded generators, fraction with weight ≤ 8, completion (global) elements, and energy-term order.

code	mech.	spins	weight p_{50}/p_{99}	≤ 8	global (weight)	order mean/p_{99}/max
surface $d = 3$	219	194	3/3	100.00%	0	2.67/16/21
surface $d = 5$	1,677	1,556	3/4	100.00%	4 (4–14)	2.82/15/24
surface $d = 7$	5,471	5,134	3/4	99.96%	6 (14–26)	2.85/14/26
surface $d = 9$	12,705	11,984	3/4	99.99%	6 (3–30)	2.85/14/27
surface $d = 11$	24,483	23,162	3/4	100.00%	13 (3–36)	2.86/14/29
surface $d = 13$	41,909	39,724	3/4	99.99%	14 (3–57)	2.87/13/26
BB $[[72, 12, 6]]$	16,164	15,726	3/4	99.88%	261 (3–104)	3.55/33/95